

Euclidean Condensate Theory (ECT): A Concise Mechanism-First Overview

Valeriy Blagovidov*

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Abstract

Euclidean Condensate Theory (ECT) studies whether Lorentzian spacetime, coherent quantum dynamics, an effective gravitational sector, and selected galactic and cosmological phenomenology can arise as sector-dependent descriptions of a single ordered phase: a real scalar medium Φ on a four-dimensional Euclidean carrier with spontaneous symmetry breaking $O(4) \rightarrow O(3)$. The central claim is architectural: structures usually postulated separately may instead appear as different regimes of one broken branch. This document is a concise mechanism-first entry point for physicists who want the main construction, the principal results, the status level of each headline claim (Level A/B/C/Open), and the most relevant tests without starting from the full preprint. The colour sector is treated separately: exact local $SU(3)$ requires the additional completion postulate P7 and does not belong to the minimal $O(4)$ -ordered condensate alone. The full preprint is available at doi:10.5281/zenodo.18917929 and the narrative companion at doi:10.5281/zenodo.19430795.

1. Introduction: from a Euclidean medium to observed physics

The purpose of Euclidean Condensate Theory (ECT) is to test whether several structures that today are introduced separately in fundamental physics can arise from one underlying ordered phase. Time, the causal cone, the Lorentzian metric, the quantum description of coherent matter, and the gravitational sector are usually layered on top of one another through independent postulates. ECT proposes that they may instead be the effective description of a single broken phase of a more primitive structure that itself contains no time, no metric, and no quantum language.

The candidate primitive structure is intentionally minimal. It is a four-dimensional Euclidean carrier on which a single real scalar medium Φ is defined. At this level the medium is pre-physical: it has no notion of energy, no Planck constant, and no preferred direction. A short list of structural properties specifies what the medium admits — connectedness, directional isotropy under four-dimensional rotations, local regularity with admissible defect submanifolds, the existence of ordered configurations, and the recognition that internal coordinates of the ordered branch are descriptive redundancies rather than physical observables.

The central structural step of ECT is the appearance of an ordered ground state in which the gradient of Φ acquires a non-vanishing expectation value along one of the four Euclidean directions. The full four-dimensional rotational symmetry is broken down to its three-dimensional subgroup;

*Correspondence: blagovidov@phystech.edu, vblagovidov@gmail.com. ORCID: 0009-0008-6707-7068.

one direction becomes distinguished. The analogy with ordering transitions in superfluid helium-3-A and with gradient-condensate phases of Bose–Einstein systems is methodological, not an identification: ECT places this kind of mechanism at the foundational layer rather than as an emergent material-physics effect on top of a pre-existing spacetime.

The broken phase carries two qualitatively distinct families of disturbances. Smooth single-valued disturbances of the collective phase form a *coherent sector*: they admit a partial reconstruction of the standard quantum formalism through a topological action scale and a reflection-positivity bridge, and they yield a recovery of Schrödinger-type evolution in a stationary coherent regime. Slowly varying disturbances of the ordered background motivate a geometric/orientation extension. The controlled equation used here is the scalar-only metric equation. A physical orientation stress $T_{\mu\nu}^{(n)}$ requires a separately specified covariant $S_n[g, n, \lambda_n]$; current anisotropic tensors are external proxies and the orientation contribution remains Open. In the galactic and cosmological applications, the same ordered branch is then extended to an infrared critical-response regime that organises the asymptotic phenomenology of galactic rotation curves and contributes to a diagnostic-level multi-probe cosmological compatibility layer.

In this picture Lorentzian spacetime, quantum coherence, and gravity appear as sector-dependent regimes of one ordered medium rather than as unrelated starting axioms. ECT is therefore organised as a common ordered-medium architecture with a separately tracked level of rigour for each sector. The colour gauge sector lies outside the minimal four-dimensional ordered structure and is introduced later through a distinct internal-completion layer.

This overview is meant to function as a working map of mechanisms, claims, status levels, and falsifiers. Its purpose is to let a reader identify quickly which parts are structural derivations, which parts depend on matching or closure assumptions, and which parts remain open. The four-level discipline introduced in §5 keeps those categories explicit throughout the document.

How to read this overview. A reader interested in the conceptual architecture should read §§1–6. A reader interested in testable phenomenology should start with §§11–14. A reader interested in possible failure modes should go directly to §§14–15. The full preprint and the narrative companion (links in §16) contain the technical derivations underlying every claim made here.

2. Scope of the framework: claims, open points, and current boundaries

The seven structural claims of ECT, stated at the level of mechanism rather than formula:

1. Lorentzian spacetime and a single universal low-energy causal cone, whose effective speed is determined by the anisotropic stiffness of the ordered branch (**Level A** for cone emergence within the broken phase; **Level B** for matching the emergent cone speed to the observed vacuum speed of light).
2. Coherent quantum sectors with a pre-quantum topological action scale identified with Planck’s constant at matching level (**Level B**), and a partial reconstruction of standard quantum formalism through reflection positivity and the Osterwalder–Schrader bridge; the spin–statistics connection holds at **Level A** for the free coherent kernel through the same Euclidean route.
3. A controlled scalar-only metric equation inside the declared ansatz, with matching to Newtonian gravity tracked separately (**Level B** within the scalar ansatz). A physical orientation stress requires covariant $S_n[g, n, \lambda_n]$ and remains **Open**; $\mathcal{X}_{\mu\nu}^{\text{cand}}$ is an external proxy only.

4. A **Level A** structural decoupling of the *direct* classical cosmological-constant baseline channel through an emergent unimodular mechanism on the ordered branch. This is a structural removal of the direct-channel catastrophe, *not* a derivation of the observed late-time value of ρ_Λ .
5. Galactic phenomenology from a specified interpolation fit (**Level C**), with the slope-four BTFR relation following algebraically only after the deep branch is assumed (**Level B/Level C**). The provisional ψ_χ one-loop result does not derive the ϕ_u AQUAL branch; the coordinate bridge, environment law, and UDG/cluster interface remain **Open** (§12).
6. A retained five-probe cosmological consistency band for a single diagnostic deformation parameter at one σ (**Level C**, presented explicitly as a multi-channel consistency diagnostic and not as a first-principles prediction).
7. A separately postulated $SU(3)$ colour-completion layer (postulate P7), introduced because exact local $SU(3)$ cannot be obtained from the minimal four-dimensional ordered condensate alone.

Current open points of ECT (tracked as **Open** unless noted otherwise):

- A strict first-principles theorem for the Born rule and the single-outcome side of the measurement problem. ECT has a **Level B** conditional route to Born-type weights via the reflection-positivity / Osterwalder–Schrader (RP/OS) Hilbert construction and a Gleason-type argument, once three reduced-description assumptions are admitted: C1, decohered alternatives carry an effective Hilbert/projector structure; C2, those alternatives are approximately orthogonal; and C3, probabilities add across mutually exclusive alternatives. Deriving C1–C3 from the bare ECT postulates, and resolving the single-outcome and measurement-update problem, remain **Open** (see §9).
- Bell-type correlations, the no-signalling condition, and the Tsirelson bound are not derived from the present ECT structure; non-factorisable coherent correlations have a structural **Level A/Level B** route, but a full Bell/CHSH derivation is **Open**.
- The full low-energy quantum-field-theoretic architecture — strict locality, cluster decomposition, analyticity, crossing, and S-matrix theory — has not been re-derived from the ordered-medium formulation; matching is at the effective-action level only.
- A first-principles derivation of the observed late-time value of the cosmological constant. ECT removes the direct vacuum-baseline channel, but the observed value remains a separate **Level C/Open** determination problem.
- A dynamical derivation of the ordered branch itself from the bare action (gradient condensation, OP-grad). In the current preprint this programme is decomposed into three recorded subproblems, and the remaining gap is localised to a single constraint-relaxation / coarse-graining bridge; the candidate bridges analysed so far do not close that gap, so the step remains **Open**.
- Derivation of hypercharge assignments, the full observed electric-charge spectrum, the flavour structure, the CKM/PMNS mixing matrices, and the three-generation hierarchy of the Standard Model.
- Solution of the strong-CP problem: the topological coupling $\Theta(\phi)F\tilde{F}$ *reopens* the problem within ECT rather than resolving it.

- A predictive QCD-like phenomenology from the colour completion: at constant background couplings the P7-layer is structurally classificatory rather than predictive.
- A predictive environment law closing the galactic interface across UDG, dwarf, and bullet-cluster matched-pair stress tests (central operational gap).
- A finished ultraviolet completion of all sectors.

ECT is therefore best read as a structured theoretical framework with a defined core, explicit matching layers, and a finite list of open dependencies. Every positive result below is presented with an explicit status badge and a locator into the full preprint.

3. Architecture of the model

ECT is built on three structural layers.

S-layer (ontology of the Φ -medium). Ten properties S1–S10 describe the Φ -medium in purely geometric and configurational terms, prior to any ordering, energy, or quantum language: connected four-dimensional Euclidean carrier (S1); $O(4)$ -directional isotropy (S2); point homogeneity (S3); local regularity with admissible defect submanifolds (S4); locality (S5); multiplicity of admissible configurations (S6); admissibility of ordered states with $\langle \partial_A \Phi \rangle \neq 0$ (S7); medium character with pointwise internal coordinates as descriptive redundancies (S8); inhomogeneity of ordering (S9); topological completeness (S10). Because the Φ -medium is pre-physical, no energy or quantum-fluctuation language is used at this layer.

Variational principle (DP). Physically realisable configurations extremise a local Euclidean functional $S_E[\Phi]$ compatible with the basic symmetries of the medium, with conditions ensuring well-posedness.

Working postulates P1–P6. The minimal concrete realisation:

- P1** Connected four-dimensional Euclidean arena $\mathcal{M}^4$, $ds_E^2 = \delta_{AB} dX^A dX^B$, with no distinguished time coordinate.
- P2** $O(4)$ -invariant fundamental action, $S_E[\Phi] = S_E[\Lambda \cdot \Phi]$ for $\Lambda \in O(4)$.
- P3** Real scalar field $\Phi(X)$ with minimal local micro-action $S_E^{\text{bare}} = \int d^4 X [\frac{1}{2}(\partial_A \Phi)^2 + V(\Phi)]$, $V = -\frac{\mu^2}{2}\Phi^2 + \frac{\lambda}{4}\Phi^4$. The double-well is a $\mathbb{Z}_2$ structure, not a “Mexican-hat”.
- P4** Physically relevant vacuum belongs to an ordered branch with $\langle \partial_A \Phi \rangle = u_0 \delta_{Aw}$, $u_0 > 0$, selecting a preferred direction w and breaking $O(4) \rightarrow O(3)$.
- P5** Pointwise internal coordinates of the ordered branch (phase $\theta(X)$, orientation $n_A(X)$) are descriptive redundancies; physical observables depend only on invariant gradient and holonomy data. The passage from P5 to the local gauge-structure route associated with $SU(2) \times U(1)$ is treated as a derived structural route on the smooth coherent branch, not as an independent postulate.
- P6** Multiplicity of admissible configurations and admissibility of non-uniform ordering (domains, transitions, defects).

Derived branch rule (BR1). On the smooth coherent sector of the ordered branch the effective order parameter is $\Phi_{\text{eff}} = \rho e^{i\theta}$, $\rho > 0$. For closed contours not intersecting defect cores,

$$\oint_C \partial_A \theta dX^A = 2\pi n, \quad n \in \mathbb{Z}, \quad (1)$$

and the elementary coherent loop introduces the pre-quantum topological action scale

$$S_{\text{loop,min}} = 2\pi S_0, \quad (2)$$

where S_0 is the ECT precursor of $\hbar$ (matching: **Level B**). BR1 is *derived*: it follows once one restricts attention to a smooth single-valued collective phase field on the broken phase.

P7 is not part of the minimal architecture. Postulate P7 (rank-3 complex internal module with Hermitian form and fixed complex volume form, yielding exact local $SU(3)$) is a separate *reverse-engineered* layer, introduced in Part IV of the preprint, and is required because $\mathfrak{su}(3) \not\subset \mathfrak{so}(4)$: the colour sector cannot reside inside the $O(4)$ ordered condensate alone. P7 is therefore additive to, not part of, the P1–P6 core.

Two energy scales and a separate length. The energy-dimension quantities are $\phi_0 \sim \bar{M}_{\text{P1}}$ and $v_2 \approx 246$ GeV. The gradient amplitude $u_0 = |\langle \partial_A \Phi \rangle|$ has dimension GeV^2 , with only the conditional dimensional estimate $u_0 \sim \phi_0 m_\sigma$. Separately, $L_{\text{gal}} = r_* = \sqrt{G_N M_{\text{bar}}/g_{\dagger}}$ is a length inside the adopted Level-C galactic functional, not a third energy amplitude; its ECT matching is Open. The logarithm $\mathcal{I}_{\text{EW}} = \ln(\phi_0/v_2) \approx 36.8$ organises the Open electroweak scale-generation programme.

4. Input economy and structural reframings

ECT spends its assumptions on a small microscopic layer (§3) and proposes structural routes to several sectors that are usually treated as independent inputs in fundamental physics. The two compact summaries below replace a long audit table: a reader who wants more detail will find it in the corresponding sections.

Input audit. ECT does not claim that fewer assumptions prove the theory; the point is to make explicit where assumptions are spent and where reconstruction is claimed.

- **Minimal ordered-medium input:** Φ -medium on $\mathcal{M}^4$, variational principle DP, working postulates P1–P6, derived branch rule BR1.
- **Recovered or structurally reconstructed:** Lorentzian causal cone; effective metric and orientation sector; coherent quantum branch; compact- $U(1)$ Abelian charge lattice; direct cosmological-constant channel decoupled.
- **Matched / induced to standard low-energy physics:** $c \equiv c_*$, $\hbar \equiv S_0$, $G_N = c_*^4/(8\pi M_G^2)$, the energy scales ϕ_0 , v_2 ; and the distinct dimension-two gradient amplitude u_0 . The galactic L_{gal} is a Level-C closure length, not a matched energy scale.
- **Closure / diagnostic layers:** the retained cosmological ε -band and late-time infrared scaling. A galactic environment/history law Ξ is an Open future hypothesis, not an implemented closure.
- **Separately postulated:** P7 (rank-3 complex internal module) for exact local $SU(3)$.
- **Open:** dynamical origin of the ordered branch itself (gradient condensation, OP-grad); full first-principles Born theorem and the single-outcome/measurement-update problem; independent derivation of the reduced-description assumptions C1–C3 (§9); Bell/CHSH and the Tsirelson bound; the full low-energy QFT architecture (locality, cluster decomposition, analyticity, crossing, S-matrix); the observed value of ρ_Λ ; hypercharge and the full Standard-Model charge/flavour/generation spectrum; the strong-CP problem within the CP-odd

topological coupling; a finished UV completion.

Structural reframings, not final solutions. ECT does not present the following items as solved at the same level. The point is to identify where the same ordered-medium architecture gives a structural reframing and where the open dependencies remain:

- **Problem of time.** Time appears as an ordered-branch direction and effective parameter on the broken phase; the full quantum-gravity problem of time is not automatically solved.
- **Quantum–classical boundary.** Coherent vs. decohered/orientation sectors as a structural criterion; the strict measurement / single-outcome problem remains **Open**.
- **Born rule. Level B** conditional RP/OS + Gleason route given assumptions C1–C3 (§9); a derivation of C1–C3 from bare P1–P6 and the single-outcome / update problem remain **Open**.
- **Cosmological-constant catastrophe.** The direct vacuum-baseline channel decouples at **Level A** through the ordered-branch fixed kinetic determinant; the observed value of ρ_Λ remains **Level C/Open**.
- **Galactic dark sector.** The adopted interpolation provides Level-C RAR/rotation-curve fits; slope-four BTFR is conditional B/C algebra inside that functional. Scalar-to-galactic matching, Ξ , and the UDG/cluster interface remain **Open**.
- **Black-hole information.** Critical-shell / reduced-state mapping onto the islands / quantum-extremal-surface language; a complete Page-curve computation within ECT remains **Open**.
- **Gauge / colour sector.** Compact- $U(1)$ Abelian lattice and chiral $SU(2)$ structural routes from the broken phase; exact local $SU(3)$ via the separate postulate P7; full SM charge/flavour/generation spectrum, confinement, and the strong-CP problem remain **Open**.

The point of this section is not to prove ECT by parsimony, nor to present a list of paradoxes resolved. It is to make explicit where ECT spends assumptions and where it claims reconstruction; the status labels remain the controlling standard.

5. Claim-status discipline

Every quantitative or qualitative result in ECT is tagged with one of four status levels. This taxonomy is used throughout the preprint and is applied uniformly below.

Level A Strict structural or theorem-level derivation within stated assumptions. Example: emergent unimodular decoupling of the direct vacuum-baseline channel; Lorentzian cone emergence within the broken phase under the stated anisotropic-stiffness conditions.

Level B Conditional or closure-level derivation; a structural route exists, but at least one matching, calibration, or effective-functional step is required to obtain a number. Example: Level-B matching $S_0 \leftrightarrow \hbar$; the BTFR slope as conditional B/C algebra inside adopted Level-C functional G; and the B/C comparison ansatz $g_0^\dagger \sim cH_0/(2\pi)$.

Level C Phenomenological or diagnostic consistency layer. Example: the retained five-probe effective band $\varepsilon \in [0.0296, 0.0376]$ at 1σ . This is empirical compatibility, not a first-principles prediction.

Open Explicitly unresolved. Example: the strict first-principles Born theorem and the single-outcome problem; the observed value of ρ_Λ ; the full Standard-Model charge, flavour, and generation structure; the strong-CP problem in the CP-odd topological sector; the environment law Ξ .

The status discipline is asymmetric on purpose: weakening a result from **Level A** to **Level B** is admissible if a derivation step is found to be incomplete; strengthening in the other direction is not permitted without an explicit rigorous derivation.

6. Derivation logic and mechanism map

The architecture of §3, the input audit of §4, and the status discipline of §5 together generate a single mechanism flow that is the subject of the remainder of this overview. Figure 1 summarises it.

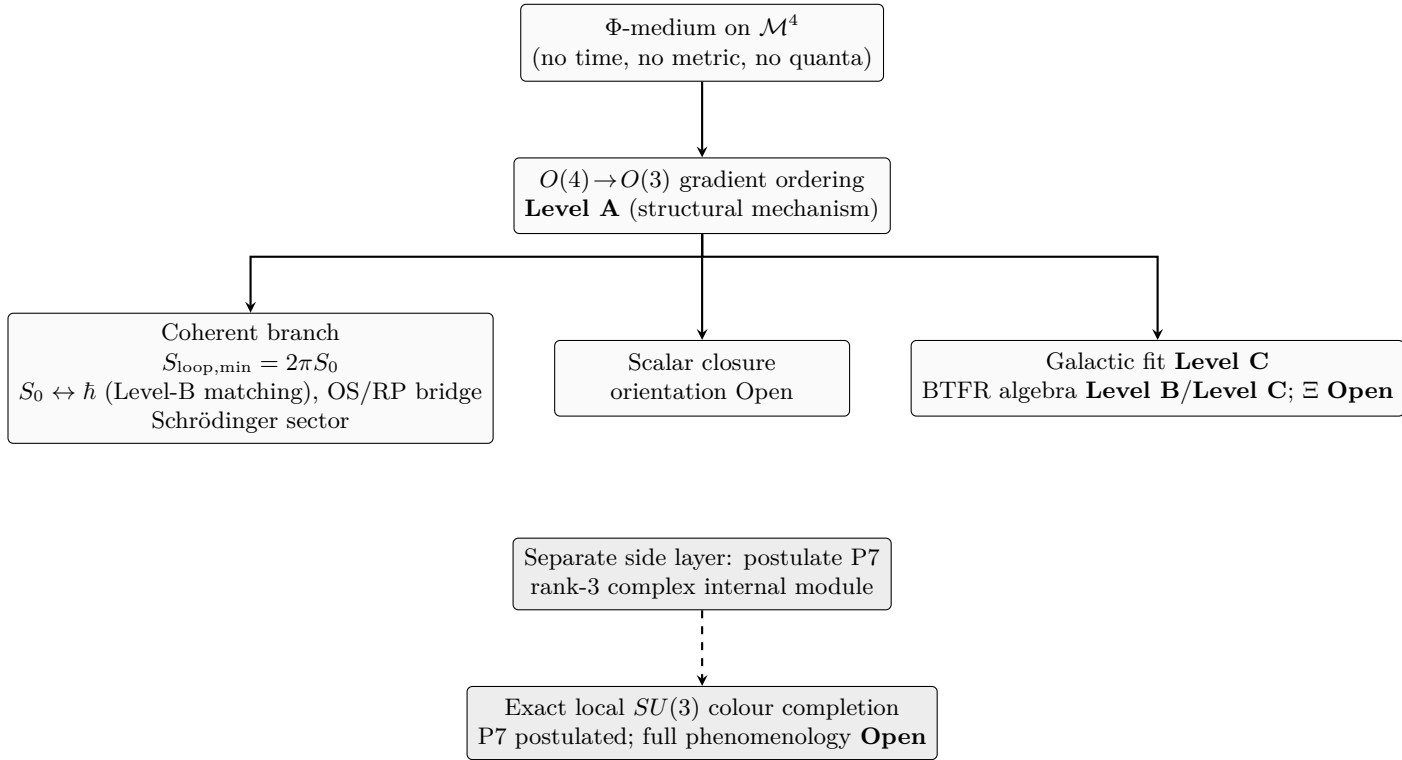


Figure 1: Simplified derivation logic of ECT. Arrows show mechanism flow, not equal-strength theorem statements. The $SU(3)$ branch is separate because it requires P7 rather than following from the minimal $O(4)$ -ordered medium; Born-type weights are part of the conditional quantum-reconstruction layer discussed in §9.

The same flow is expressed verbally as five structural routes from one medium to qualitatively distinct phenomena:

1. **Ordered branch** (P4, anisotropic kinetic tensor with $\alpha > \beta$): selection of a Lorentzian causal cone and an effective signature $(-, +, +, +)$. §8.
2. **Smooth coherent sector** (BR1, smooth single-valued θ): integer winding, $S_{\text{loop},\text{min}} = 2\pi S_0$, reflection positivity, and an Osterwalder–Schrader bridge yielding a *partial Hilbert-space reconstruction route* for the coherent quantum sector. §9.

3. **Scalar metric response and Open orientation extension:** the displayed equation is scalar-only. A physical $T_{\mu\nu}^{(n)}$ requires covariant $S_n[g, n, \lambda_n]$; $\mathcal{X}_{\mu\nu}^{\text{cand}}$ is only an external proxy. §10.
4. **Specified galactic interpolation:** Level-C RAR/rotation fits and conditional BTFR algebra. It is not derived by the provisional ψ_χ loop or by the scalar trace equation. A separate scalar-only diagnostic closure is used for cosmology. §§11–12.
5. **Reverse-engineered colour layer** (P7, rank-3 complex internal module): exact local $SU(3)$ at the cost of a separate postulate beyond the $O(4)$ -ordered condensate. §13.

7. How ECT differs from nearby frameworks

ECT is not a recombination of existing programmes. The two tables below locate ECT against the closest established or proposed frameworks; the split into *conceptual foundations* (Table 1) and *phenomenology and gauge sector* (Table 2) is for readability only.

Table 1: Conceptual foundations.

Sector	Standard approach	ap-	Nearby alternatives	ECT mechanism	Status
Spacetime signature	Lorentzian metric assumed		CDT [1], causal sets, analogue gravity; clock-field route [7]	Ordered Euclidean branch selects an effective Lorentzian cone; formula and matching given in §8.	Level A existence; Level B matching
Quantum formalism	Hilbert space and Born rule postulated		Many-worlds, Bohm, QBism, OS reconstruction	Coherent branch gives an RP/OS route to Hilbert-space dynamics; Born-type weights remain conditional.	Level B structural; Born thm. Open
Gravity	Metric field fundamental		Sakharov-induced gravity, Jacobson thermodynamic derivation [5], Einstein-aether/vector-tensor analogues [4], Verlinde entropic gravity [10], analogue gravity [11, 8]	Scalar metric ansatz; orientation Open.	Level B; Open
Cosmological constant	Vacuum energy gravitates; demands cancellation $\sim 10^{120}$		Unimodular gravity, sequestering, Weinberg no-go arguments	Direct vacuum-baseline channel decouples through the ordered-branch fixed determinant; observed value remains diagnostic / open.	Level A direct; observed ρ_Λ Level C/Open

Related independent work on Lorentz-signature emergence. Mukohyama and Uzan studied the emergence of Lorentzian dynamics from a four-dimensional Riemannian microscopic structure through a clock-field derivative background [7]. This is nearby prior work on emergent

Lorentz signature, but it is not the basis of ECT: the present construction uses a different scalar-medium architecture, an ordered-gradient condensate, a coherent/decohered branch split, and a different gravitational and cosmological programme.

Table 2: Phenomenology and gauge sector.

Sector		Standard proach	ap- proach	Nearby alterna- tives	ECT mechanism	Status
Galactic sector	dark	Particle dark matter, NFW halos		MOND/AQUAL [6, 3], Verlinde emergent dark gravity	Specified interpolation fit; BTFR slope 4 is conditional algebra inside its deep branch (§12).	fit Level C ; slope Level B/Level C ; Ξ Open
Hubble / S_8 tensions		Parameter extensions to Λ CDM, EDE, modified gravity		EDE, interacting DE, $f(R)$ [9], quintessence [2]	Single late-time deformation parameter ε ; retained five-probe consistency band at 1σ (§11).	Level C consistency; first-principles Open
Gauge sector / $SU(3)$		Group assigned by the Standard Model		GUTs, internal-geometry programmes	Compact- $U(1)$ Abelian charge lattice (derived); chiral $SU(2)$ from the oriented broken phase; exact local $SU(3)$ via the separate postulate P7.	$U(1)$ Level A/Level B ; $SU(2)$ Level B ; $SU(3)$ via P7; full SM spectrum Open

Three further differences merit explicit statement: ECT *does not* assume the existence of Lorentzian spacetime anywhere as a fundamental ingredient; the specified galactic fit contains no particle-halo term, without excluding hidden-sector particles; the late-time closure likewise does not take dark energy as an external fluid; and ECT *does not* resolve the strong-CP problem.

8. Emergent spacetime and the universal cone

In the broken phase, linearised excitations around the ordered background propagate with an effective kinetic tensor that becomes anisotropic in the direction w selected by $\langle \partial_A \Phi \rangle$. For canonical anisotropic stiffness couplings (α, β) the longitudinal/transverse split admits a Lorentzian signature window if and only if $\alpha > \beta$, in which case the effective low-energy cone has a single universal propagation speed:

$$c_* = \sqrt{\beta/(\alpha - \beta)}, \quad \text{Lorentzian window: } \alpha > \beta. \quad (3)$$

Mechanism in ECT. $O(4) \rightarrow O(3)$ SSB of the gradient condensate selects a preferred direction; the kinematic constraint $n^A n_A = 1$ on the orientation variable, combined with $\alpha > \beta$, makes the longitudinal mode disperse on a single emergent light cone. Matching of c_* to the observed vacuum speed c is a separate calibration step.

What this establishes. A concrete route from Euclidean ordering to Lorentzian signature, an effective time direction, and a universal low-energy causal cone within the broken phase.

Status. **Level A** for emergence of the cone within the broken phase; **Level B** for identifying

c_* with the observed c .

Where in the full preprint. Part I, ordered-branch and metric-classification sections; cone universality theorem.

Falsifier. An unsuppressed sector-dependent low-energy propagation speed (e.g. photon–graviton differences beyond the GW170817 bound) would falsify universal-cone emergence.

9. Quantum sector: what is recovered and what remains open

ECT reconstructs the quantum sector on the coherent branch rather than postulating Hilbert space at the foundational level. The smooth coherent sector of the ordered branch supports a partial reconstruction of the standard quantum formalism through the Euclidean path-integral route, with reflection positivity providing the Osterwalder–Schrader bridge to a Hilbert space and a unitary evolution on the coherent stationary sector. At the overview level the coherent-sector evolution is written schematically as

$$iS_0 \partial_t \Psi = \hat{H}_{\text{coh}} \Psi + \Delta_{\text{dec}} \Psi + \Delta_{\nabla} \Psi, \quad (4)$$

where Δ_{dec} denotes reduced-state / decoherence corrections from the influence functional and Δ_{∇} collects higher-gradient corrections to the kinetic operator; the explicit form of these correction operators is given in the corresponding part of the full preprint and is not needed at the overview level. For coherent stationary eigenmodes $\hat{H}_{\text{coh}} \Psi = E \Psi$, $\Psi \sim e^{-i\omega t}$, this yields $E = S_0 \omega$ up to the controlled corrections. The standard Planck–Einstein relation $E = \hbar \omega$ then follows from the **Level B** matching $S_0 = \hbar$.

Born-type weights as a conditional reconstruction. ECT does not insert Born weights by fiat. Instead, once the reduced coherent/decohered description reaches three specific assumptions — C1: decohered alternatives admit an effective Hilbert/projector structure; C2: those alternatives are approximately orthogonal; C3: probabilities add across mutually exclusive alternatives — the RP/OS Hilbert construction together with a Gleason-type uniqueness argument selects the Born weights. This conditional reconstruction has **Level B** status. What remains open is earlier in the chain: deriving C1–C3 directly from the bare ECT postulates, obtaining a first-principles Born theorem without these reduced-description assumptions, and resolving the single-outcome / measurement-update problem. Common-medium nonfactorisation is a candidate **Level B/Open** route; the physical state/channel, Bell and no-signalling correlators, and Tsirelson saturation remain **Open**.

The spin–statistics connection is established at **Level A** for the free coherent kernel through the Euclidean Osterwalder–Schrader route (a Lüscher–Osterwalder-class argument verified against the original sources); the interacting completion remains **Open**. The free antisymmetric kernel gives an exact mathematical Pauli/CAR statement; applying it to physical interacting ECT matter remains **Level B/Open** pending the spinor/kernel completion. A conditional CPT route is identified through the Euclidean Osterwalder–Schrader framework; a preferred-direction operator arises only as a CPT-odd branch-sensitive operator consistent with the exact symmetry backbone. PES-R provides a status-separated inventory of stationarity, pairwise record, finite-time stability, and protocol diagnostics; it is not a universal persistent-sector selection theorem. PES is retained as a Level-B organising/filter framework, not as a universal one-history action. Its operational objects must remain separate: the pairwise record/decoherence exponent $\Phi_{ab}(T)$, a declared translation-susceptibility diagnostic $\Gamma_{\text{tr}}[q; \eta]$, the decay rate/width γ_a (or $S_0 \gamma_a$ in action units), and protocol entropy production Σ . None of these supplies a general mean-EOM variational additive action-

plus-decoherence mean-EOM functional, and no universal forward/reverse probability law follows without a named protocol. The ECT record-sector vertex, state, and noise kernel remain Open/Not Identifiable.

Mechanism in ECT. Reflection positivity and OS reconstruction on the coherent branch, together with the topological pre-quantum scale $S_{\text{loop,min}} = 2\pi S_0$.

What this establishes. A coherent-sector route to Hilbert-space dynamics, the Schrödinger sector, and a topological precursor of $\hbar$; Born-type weights follow once the reduced-description assumptions C1–C3 are in place.

Status. **Level B** for the OS bridge, PES, and the $S_0 = \hbar$ matching; **Level B** conditional for Born-type weights once assumptions C1–C3 are admitted; **Open** for the first-principles Born theorem from bare P1–P6, for C1–C3 themselves, and for the single-outcome / measurement-update problem; **Level A** for the spin–statistics connection on the free coherent kernel (interacting completion **Open**).

Where in the full preprint. Part III, sections on coherent dynamics, PES, hybrid consistency, and the influence-functional decoherence model, with the separately reviewed PES-R repair still pending before the record-sector owner blocks become current.

Discriminant. Robust violation of the explicitly matched coherent-sector quantum dynamics would constrain that closure. Present gravity-decoherence and mediator-loophole bounds test external comparator classes, not an absolute ECT rate: the ECT record vertex, state, and noise kernel are Open/Not Identifiable.

DP-02 negative spectral result. The tested minimal scalar vacuum channel has $J_\phi \propto \omega^3$; the equal-mass Gaussian composite vacuum channel has $J_\mathcal{O} \propto \omega^9$ with a finite-time T^{-8} envelope; and its thermal regime has $J_\mathcal{O}^{\text{th}} \propto T_b^5/k$. None gives the universal Diósi–Penrose $1/k^2$ infrared form. Alternative ECT state/pole/vertex constructions remain Open.

10. Gravity and the cosmological-constant catastrophe

The controlled metric equation in the displayed scalar-only Jordan-frame ansatz is, in an undivided convention,

$$F G_{\mu\nu} = \frac{8\pi G_N}{c^4} T_{\mu\nu} + \mathcal{T}_{\mu\nu}^{(q)}, \quad (5)$$

where $\mathcal{T}_{\mu\nu}^{(q)}$ collects scalar derivative and potential terms. The factor G_N/F appears only after all terms are divided by F . The displayed action contains no independent orientation field and therefore does not derive $T_{\mu\nu}^{(n)}$; $\mathcal{X}_{\mu\nu}^{\text{cand}}$ is only an external proxy. A covariant $S_n[g, n, \lambda_n]$ and its metric variation remain **Open**. Dividing the equation by F introduces no orientation term. The scalar audit uses $a = f_{,q}/f$, $\mathcal{A} = K + 3f_{,q}^2/(2f)$, and $\mathcal{E} = V_{,q} - 2aV - (a/2)\rho_m$; at a stable homogeneous root, $d \ln \bar{f}/d\rho_m = a^2/(2\mathcal{E}_{,q}) \geq 0$ and $m_{\text{loc}}^2 = \mathcal{E}_{,q}/\mathcal{A}$ are conditional local-response statements, not finite-body screening. The FRW/Hubble/JWST diagnostics use the scalar-only background. The Planck-scale matching $G_N = c_*^4/(8\pi M_G^2)$ is kept as a separate matching relation.

M (the microscopic director operator), S (this Level-B scalar ansatz), and G (the adopted Level-C galactic functional) are distinct. $M \neq S$ by operator content and $S \rightarrow G$ is Open. The historical LOF appendix is a mixed branch audit: the heavy-radial proxy result does not select another coordinate or the galactic functional, and the matter-to-Lorentz-order map remains Open.

The cosmological-constant catastrophe (direct channel). In a standard general-relativistic framework, the quartic double-well baseline $V(\phi_0) = -\mu^4/(4\lambda)$ would gravitate as a cosmological

constant of order $\lambda M_{\text{Pl}}^4/4$, overshooting $\rho_\Lambda^{\text{obs}} \sim (2.3 \times 10^{-3} \text{ eV})^4$ by $\sim 10^{120}$. In ECT this catastrophe is removed at **Level A** for the leading ordered-branch sector through an *emergent unimodular decoupling* theorem: the determinant of the ordered-branch kinetic tensor is fixed by the kinematic constraint $n^A n_A = 1$:

$$\sqrt{-\det K^{AB}} = \beta^2/c_*, \quad (6)$$

and consequently the bare baseline $V(\phi_0)$ does not source the local emergent gravitational dynamics. Conditional on a one-line spectral assumption on the full quadratic fluctuation operator (condition H_Λ), the zero-derivative one-loop vacuum contribution is absorbed into the same fixed kinetic-determinant structure. The direct 10^{120} -level UV-to-IR cancellation characteristic of the standard framework is therefore structurally absent in the leading ordered-branch sector. Two further strengthenings are recorded in the current preprint: the decoupling statement is realised in an explicit exact FRW ordered-branch background (the realisation-side tasks of the cosmological-constant programme are completed), and an area-law holographic entropy bound is established as a structural property of the ordered medium (**Level A** for the scaling, **Level B** for the coefficient).

Remaining determination problem. The observed numerical value of ρ_Λ remains a separate **Level C** determination problem; the natural infrared scaling $\rho_\Lambda \sim c_\Lambda M_{\text{Pl}}^2 H_0^2$ with $c_\Lambda = \mathcal{O}(1)$ is order-of-magnitude consistent but the coefficient is not yet derived.

Mechanism in ECT. Scalar-only metric response in the declared Jordan-frame ansatz; fixed determinant of the ordered-branch kinetic tensor through $n^A n_A = 1$ for the separate direct-channel decoupling result.

What this establishes. Consistent scalar-only metric normalisation and a direct-channel decoupling of the vacuum baseline in the leading ordered branch. A covariant orientation-stress source is not established by the displayed action.

Status. exact algebra within the declared Level-B scalar ansatz for the metric normalisation; separately **Level A** for the direct-channel decoupling theorem; **Open** for the orientation-stress extension; **Level C** for the observed value of ρ_Λ ; **Open** for the first-principles value of c_Λ .

Where in the full preprint. Part II, sections on generalised Einstein equations, the cosmological-constant problem, and the decoupling theorem.

Falsifier. A robust demonstration that constant vacuum-energy shifts source the local emergent gravitational equations with standard GR strength would falsify the direct-channel decoupling claim. Independently, a statistically robust phantom crossing $w < -1$ would put the present ordered-branch dark-energy closure under serious pressure.

11. Cosmology and current observational tensions

The macroscopic ordered-branch closure admits a uniform effective late-time deformation,

$$G_{\text{eff}}(z)/G_N = (1+z)^{2\varepsilon}, \quad (7)$$

where ε is a single diagnostic parameter of the *retained five-probe analysis layer* (Hubble plus sound horizon r_s , JWST high- z extraction, cosmic chronometers, $f\sigma_8$ redshift-space distortions, integrated Sachs–Wolfe cross-correlation). A common effective constraint from the five retained probes yields

$$\varepsilon \in [+0.0296, +0.0376] \quad (1\sigma), \quad [+0.021, +0.042] \quad (2\sigma). \quad (8)$$

This is a *consistency window* across observationally distinct channels with different kernels and systematics. It should be read as an indirect multi-channel *viability test* of the present effective parameterisation, rather than yet as a global likelihood combination or a closed first-principles derivation.

The associated age indicators of the Lorentzian branch are ~ 13.5 Gyr under baseline- H_0 mapping and ~ 12.5 Gyr under inferred- H_0 mapping; the JWST early-galaxy sector contributes the binding lower edge of the retained band under the present extraction pipeline. Excluded probes are listed separately: BAO and lensing A_{lens} as methodology-limited under the uniform- ε ansatz; S_8 (KiDS-1000) and σ_8 from clusters as outside the present ε -sector under a category-error interpretation; the SN Ia channel as non-independent of the retained anchors.

Within the same ordered-branch background, the late-time dark-energy sector is a *thawing* closure. The exact background relation is $w_0 = (r_0 - 1)/(r_0 + 1)$, where r_0 is the present kinetic-to-condensate energy ratio ($r_0 \approx 0.095$ at the calibrated benchmark), with the linearised form $w(z) \simeq -1 + 2f_0 e^{-\kappa z}$, $f_0 \approx 0.087$. The frequently quoted value $w_0 \approx -0.83$ is a DR1-anchored *calibrated thawing benchmark* (**Level B**): a useful reference point rather than a derivation from the microscopic layer. The closure keeps $w(z) \geq -1$ at all redshifts.

Mechanism in ECT. A single ordered-branch deformation parameter ε tracking the same condensate background that controls both the dark-energy sector and the epoch-dependent baseline acceleration scale $g_{\text{bg}}^\dagger(z) = c_* H(z)/(2\pi)$.

What this establishes. A retained late-time consistency window in which one effective deformation parameter organises several observational channels at once.

Status. **Level C** consistency layer; first-principles derivation of $\varepsilon(z)$ from the closure functions is **Open**.

Where in the full preprint. Part II, retained five-probe section and its methodological appendix.

Falsifier. A future joint analysis with the same retained probes that stably pulls ε below zero (in particular through JWST high- z extraction stabilising at a binding lower edge $\varepsilon \leq 0$) would put the present sector under serious pressure.

12. Galactic phenomenology

In the deep low-acceleration regime of the adopted galactic closure the asymptotic galactic field equation reduces to $g \simeq \sqrt{g_N g_\dagger}$, and the asymptotically flat rotation-curve velocity satisfies

$$v_\infty^4 = G_N M_b g_\dagger. \quad (9)$$

The asymptotic baryonic Tully–Fisher slope of 4 is algebraic only inside this specified deep-branch interpolation; the closure itself is not derived from the corrected scalar action. In particular, $\phi_u = \beta_\phi^{-1} \ln(u/u_\infty)$ is distinct from $\psi_\chi = \ln(\chi/\chi_{\text{vac}})$, and the provisional $K_\chi \propto e^{-\psi_\chi/2}$ loop does not derive the AQUAL branch. The practical fit returns per-galaxy $g_{\text{eff}}^\dagger$ values (Level C). The reference $g_0^\dagger \approx 1.04 \times 10^{-10} \text{ m/s}^2$ is a matched B/C benchmark, compared with the empirical $a_0 \approx 1.2 \times 10^{-10} \text{ m/s}^2$; it is not a universal fit output. A possible regression $g_{\text{eff}}^\dagger/g_0^\dagger = \Xi(\rho_{\text{env}}, \bar{\phi}_{\text{env}}, \dots)$ is an Open future hypothesis, not the current implementation. The sample medians are $\chi_r^2 \approx 2.7$ (fixed M/L) and ≈ 1.4 (free M/L), versus the quoted MOND comparator ≈ 3.1 on comparable footing. The conditional length $L_{\text{gal}} = r_* = \sqrt{G_N M_b / g^\dagger}$ belongs to G and is not a condensate energy scale. The physical S→G map, particle content, and cosmological fraction budget remain

Open.

The UDG inverse diagnostic now uses the Wolf coefficient 3 at $R_{1/2} = 4R_e/3$ and the unconstrained quantity $\Xi_{\text{alg}} = \mathcal{R}_{\text{dyn}}(\mathcal{R}_{\text{dyn}} - 1)g_N/g_0^\dagger$. It is a physical environment factor only when $\Xi_{\text{alg}} \geq 0$. Central values are -0.00369 (DF4; no admissible nonnegative- Ξ solution), 0.00516 (FCC 224), 0.892 (NGC 5846-UDG1), and 4.981 (Dragonfly 44); DF2’s dispersion bracket gives $0.031\text{--}0.423$. The previous logarithmic nuisance-parameter band is withdrawn; a full Jeans posterior is Open.

The displayed force-dimensionality curve belongs to the separate illustrative point-source law $\mu_{\text{dim}} = x/\sqrt{1+x^2}$, not practical G; an extended-galaxy extraction requires measured $p_N(r)$ and has not been performed. The cluster script is only an exact order/argmax-preservation identity for hand-set maps. It supplies no morphology/lensing evidence or observed amplitude inference; a physical cluster response and likelihood remain Open.

Mechanism in ECT. Adopted low-acceleration interpolation closure; BTFR slope as an algebraic corollary of its deep-branch limit; exploratory future regression of per-galaxy $g_{\text{eff}}^\dagger$ through a prespecified Open law Ξ ; the current fits do not implement it.

What this establishes. A Level C fit benchmark and a conditional algebraic BTFR slope inside the adopted closure; it does not establish the physical $u \leftrightarrow \chi$ bridge or scalar screening.

Status. BTFR slope **Level B/Level C** conditional on the closure; per-galaxy $g_{\text{eff}}^\dagger$ fit outputs **Level C**; matched $g_0^\dagger$ reference **Level B/Level C**; the environment/history law Ξ , UDG closure, and finite-body screening **Open**.

Where in the full preprint. Part II, sections on galactic dynamics, radial acceleration relation, SPARC fits, and the UDG stress-test appendix.

Discriminant. A robust BTFR slope inconsistent with 4 would reject the adopted deep-branch closure. Independently, a prespecified boundary-aware Jeans likelihood for the declared UDG sample tests that closure; failure rejects the closure, not ECT as a whole. An unrestricted Ξ family is not identifiable and is therefore not used as a falsifier.

13. Gauge sector and the $SU(3)$ completion

On the smooth coherent branch the compact phase θ admits a bundle covariance whose connection plays the role of an effective $U(1)$ gauge field; integer winding around defect cores then implies a compact- $U(1)$ Abelian charge lattice. The full observed Standard-Model charge spectrum — including hypercharge, flavour, and generation structure — is not derived from this mechanism and remains **Open**. A recorded *colour-interlock* consistency layer now constrains completions: with the standard $\mathbb{Z}_6$ -type global quotient, hypercharge, colour and weak data must satisfy an explicit congruence, which fixes the fractional electric pattern per colour class ($Q \equiv -t/3 \pmod{1}$) and reduces the residual freedom to an explicitly characterised discrete family — a consistency requirement on completions, not a derivation. A chiral $SU(2)$ sector is structurally linked to the oriented broken phase through the orientation variable n_A and is the subject of the layered electroweak programme (OP-EW-scale, OP-EW-locking, OP-EW-gauge, OP-EW-naturalness, OP-EW-matter). The electroweak anomaly conditions of the emergent chiral sector are verified with the inherited matter content (**Level A**), and the preferred-direction coupling is transparent to chiral anomaly coefficients.

The colour sector is structurally distinct. The algebraic obstruction $\mathfrak{su}(3) \not\subset \mathfrak{so}(4)$ shows that exact local $SU(3)$ cannot arise from the minimal $O(4)$ -ordered condensate alone. ECT therefore introduces, as a separate reverse-engineered layer:

Postulate P7. The Φ -medium is equipped with a local rank-3 complex internal module, a Hermitian form, and a fixed complex volume form. The corresponding frame redundancy is $U(3)$; the fixed complex volume form reduces it to $SU(3)$.

Three obstruction theorems are then proved in Part IV: no tree-level gluon mass, no Higgs-colour route, and the oscillator tower of the colour sector is not the physical colour spectrum. At the anomaly level, cancellation conditions alone do not select the colour rank (they require only an odd number of colours), so the rank-three input of P7 must come from inherited charge data or from the open zero-mode programme. A central negative result is established: at constant background couplings, the P7 colour-sector completion is structurally *classificatory rather than predictive*; it does not generate independent predictive strong-sector physics beyond a redefinition of matching conventions. The topological coupling $\Theta(\phi)F\tilde{F}$ *reopens* the strong-CP problem within ECT rather than resolving it. For the Yang–Mills mass gap, Part IV separates the physical question from the mathematical Clay problem: the Clay continuum object is shown not to be a fundamental referent of ECT (**Level A**). The physical colour gap is instead formulated as an ECT-native finite-cutoff problem. In the current heat-kernel/HKLoc formulation, the polymer-entry pair is reduced to a delayed-entry sufficient criterion inside the chosen blocking framework: compact positive-class preservation and local activity bookkeeping are closed at the stated framework level, while the remaining one-sided anisotropy budget is satisfied in the existence-over-blocks form for finite bounded plane anisotropy with $1 + \zeta_Y > 0$ (**Level B**). This remains a conditional ECT-native gap statement, not a Clay-continuum Yang–Mills construction.

Mechanism in ECT. Compact- $U(1)$ Abelian charge lattice from the bundle covariance of the compact phase; chiral $SU(2)$ from the oriented broken phase; exact local $SU(3)$ only via the additional P7 internal-module postulate.

What this establishes. A derived compact- $U(1)$ lattice, a structural chiral $SU(2)$ route, and a clearly separated P7 completion for exact local $SU(3)$.

Status. Compact- $U(1)$ charge lattice **Level A/Level B** on the smooth coherent branch; chiral $SU(2)$ structural **Level B**; $SU(3)$ postulated (P7); full SM charge/flavour/generation spectrum, confinement, and strong CP **Open**.

Where in the full preprint. Part I (compact phase, $U(1)$, non-Abelian embedding), Part IV ($SU(3)$ completion, P7, obstruction theorems).

Falsifier. A confirmed low-energy colour-sector effect requiring a structure incompatible with the P7 internal-module construction or with exact local $SU(3)$ in the ECT completion would falsify the present colour-sector layer.

14. Falsifiers and discriminants

The preprint maintains an explicit registry of falsifiers and a second registry of future discriminants ND1–ND13 in its prediction and discriminant tables. The most informative falsifiers and discriminants of the present state of ECT are:

1. **F1 — exploratory environment/history dependence of $g^\dagger$.** The fitted scale $g_{\text{eff}}^\dagger$ may be regressed against declared environment and history covariates as a test of the Open law Ξ . No sign or amplitude is currently predicted; a null result rejects only the tested Ξ ansatz, while a correlation does not derive ECT.
2. **GW-speed / Lorentz-cone universality.** The present effective parametrisation is compatible with the GW170817 bound $|c_{\text{GW}}/c - 1| < 6 \times 10^{-15}$; the Planck-suppressed GW delay

estimate is $\Delta t_{\text{GW}} \sim 10^{-64}$ s. A robust sector-dependent low-energy speed beyond this bound would falsify universal-cone emergence.

3. **ND-13 — boundary-aware UDG inverse test.** DF4 has $\Xi_{\text{alg}} < 0$ at its central Wolf-proxy inputs and hence no physical $\Xi \geq 0$ solution inside the adopted closure. A full Jeans likelihood tests that closure, not ECT as a whole.
4. **Gravity-induced entanglement (BMV-class).** An operational criterion separates the ϕ -channel-distinguishability regime (no GIE) from the orientation-based candidate entangling regime.
5. **Gravitational record channel.** Current mesoscopic values, including 38 ms for a micron nanosphere, are imported Diósi–Penrose benchmarks. The ECT physical vertex, state-dependent noise kernel, finite-body charge, and absolute rate remain Open/Not Identifiable. Legacy crossover rows labelled only by (P, T) are not reproducible under their stated ideal-gas N_2 convention.
6. **No phantom crossing.** The ordered-branch closure gives $w(z) \geq -1$ at all redshifts in the present sector; a statistically robust $w < -1$ detection would pressure the closure.
7. **Negative discriminator: S_8 / weak-lensing sector.** The current ε -mechanism is not designed to relax the S_8 tension; this sector is treated as a negative discriminator of the present effective layer.

Further discriminants — including gravitational memory and soft-graviton structure, black-hole Page-curve-compatible evaporation, and defect-core localised excitations — are catalogued in the full preprint.

ECT currently treats any additional excitations referenced above as candidate excitation channels of the Φ -medium rather than as established new-particle predictions beyond the known degrees of freedom.

15. Summary status map

Table 3: ECT summary status map. “Open dependency” lists the next falsifier or open dependency that could weaken the row’s status. Section pointers are given in the corresponding mechanism boxes; full technical locations are in the preprint.

Result	Mechanism	Status	Open dependency
Lorentzian cone	Ordered-branch SSB; $\alpha > \beta$	Level A/Level B	GW170817-class sector-dependence
Coherent quantum sector; $S_0 \leftrightarrow \hbar$ is Level-B matching	$S_{\text{loop}, \min} = 2\pi S_0$; OS/RP bridge	Level B	Independent small-loop action standard
Scalar metric response / orientation extension	Scalar ansatz; external proxy	Level B; Open	Covariant S_n ; $T_{\mu\nu}^{(n)}$
CC direct-channel decoupling	Fixed kinetic determinant from $n^A n_A = 1$	Level A	Spectral condition on the fluctuation operator
Galactic RAR / BTFR	Adopted interpolation; conditional deep-branch algebra	Level C fit / Level B/Level C slope	Physical coordinate bridge; branch selection

Table 3, continued.

Result	Mechanism	Status	Open dependency
Scalar screening / environment law	Local trace response; exploratory Ξ	Open	Body charge + PPN profile; sign/amplitude of Ξ
Diósi–Penrose benchmark / ECT record channel	Imported D–P functional; ECT kernel not identified	Level C/Open	Physical projector, state, noise and normalisation
Cosmological ε -band	Uniform effective late-time deformation	Level C	JWST extraction at $\varepsilon \leq 0$
Born-type weights / measurement	RP/OS Hilbert + Gleason; C1–C3 admitted	Level B conditional / Open	Derivation of C1–C3; single-outcome update
Spin–statistics (free coherent kernel)	Euclidean OS route (Lüscher–OS class)	Level A free kernel	Interacting completion
Gauge sector / P7	Compact- $U(1)$, chiral $SU(2)$ route; $SU(3)$ via P7	A/B for $U(1)$; B for $SU(2)$; P7 for $SU(3)$	Full SM spectrum (interlock-constrained discrete family); confinement; strong CP

Additional status entries for the Pauli exclusion principle, the observed value of ρ_Λ , the structural normalisation of $g_0^\dagger$, and the matching of the low-energy quantum-field-theoretic architecture are tracked in the full preprint.

16. Resources and invitation to engage

Full technical preprint. *Euclidean Condensate Theory (ECT): Emergence of Spacetime, Quantum Mechanics, and Gravity from Spontaneous $O(4)$ Symmetry Breaking*, V. Blagovidov, 2026, 784 pp. in the frozen Round 58 proposal build (page count may change before release).
doi:10.5281/zenodo.18917929

Narrative companion (conceptual guide to the full preprint), ~ 104 pp.
doi:10.5281/zenodo.19430795

Code and figure-generation scripts.

github.com/chufelo/ECT-preprint-code (SPARC fits, retained ε -band analysis, UDG stress test, condensate-scale plots, comparative timeline).

Author contact. Valeriy Blagovidov, blagovidov@phystech.edu, vblagovidov@gmail.com. ORCID: 0009-0008-6707-7068.

Invitation. The most useful engagement, for a skeptical reader of this document, is to identify whether any of the seven structural claims of §2 (i) contradict an established result, (ii) admit a simpler mechanism in an existing framework, or (iii) depend on a hidden auxiliary assumption not flagged here. Specific feedback on the explicitly open items listed in §2 — the strict first-principles Born theorem and the single-outcome problem, the Bell/CHSH/Tsirelson status, the observed value of ρ_Λ , the environment law Ξ , the dynamical origin of the ordered branch (OP-grad), the hypercharge and full Standard-Model charge/flavour/generation spectrum, and the strong-CP problem — is especially welcome.

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